VLSI Devices
Lecture 8

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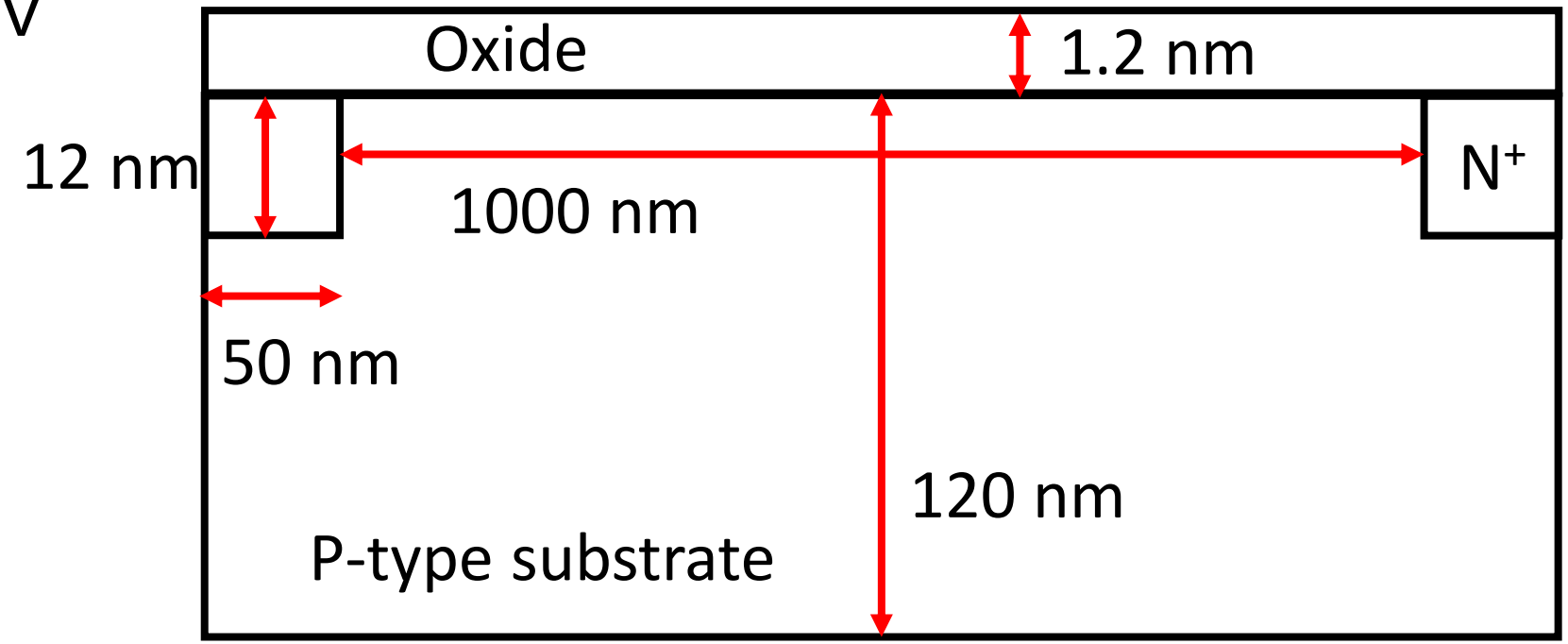
L8

Example of a long-channel MOSFET

- Effective oxide thickness of 1.2 nm
  - Gate workfunction of 4.3 eV
  - Substrate doping of $1.5X10^{18}$ cm^{-3}
  - V_{dd} of 1.2 V

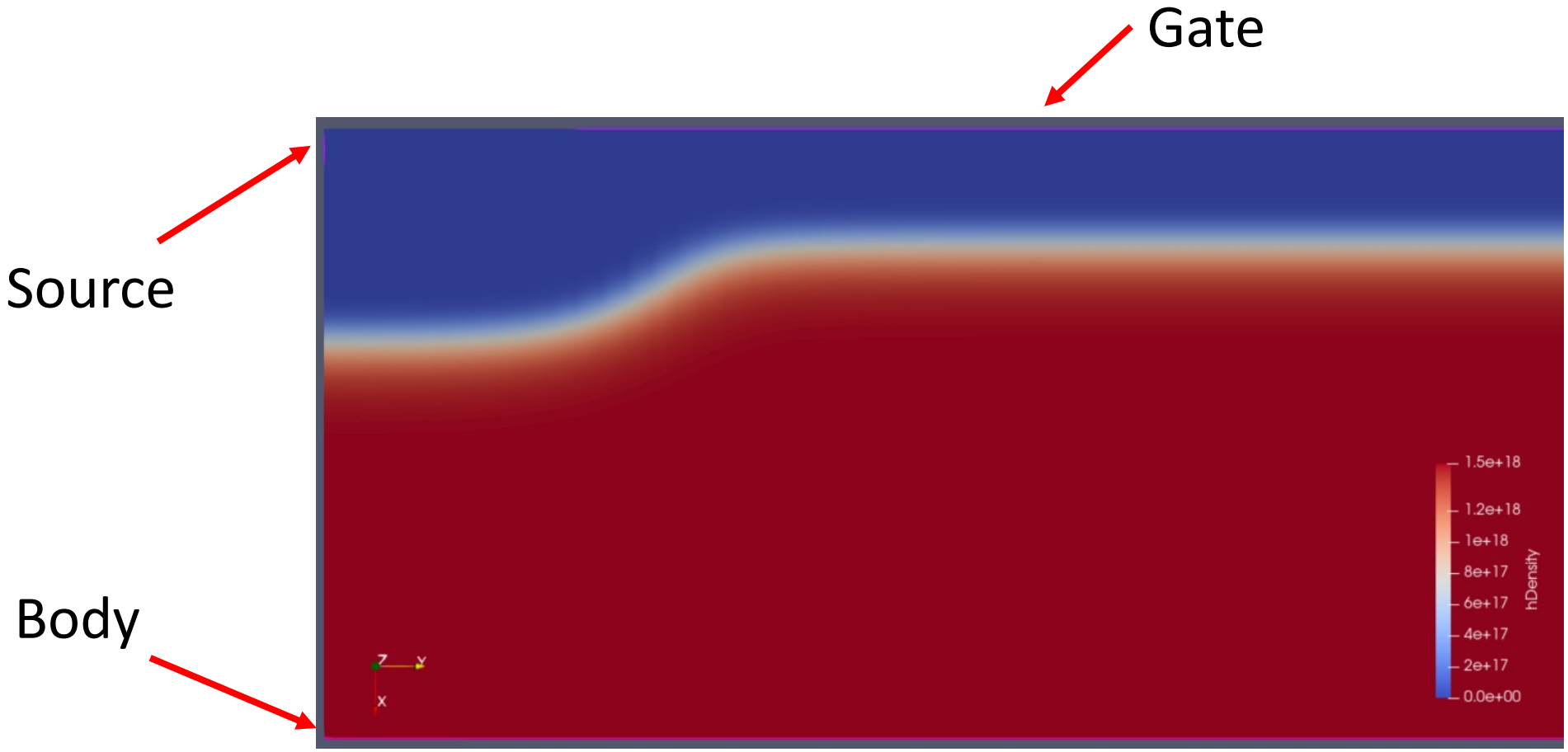


Computer program

- For accurate prediction of electrical characteristics
  - Semiconductor device simulator

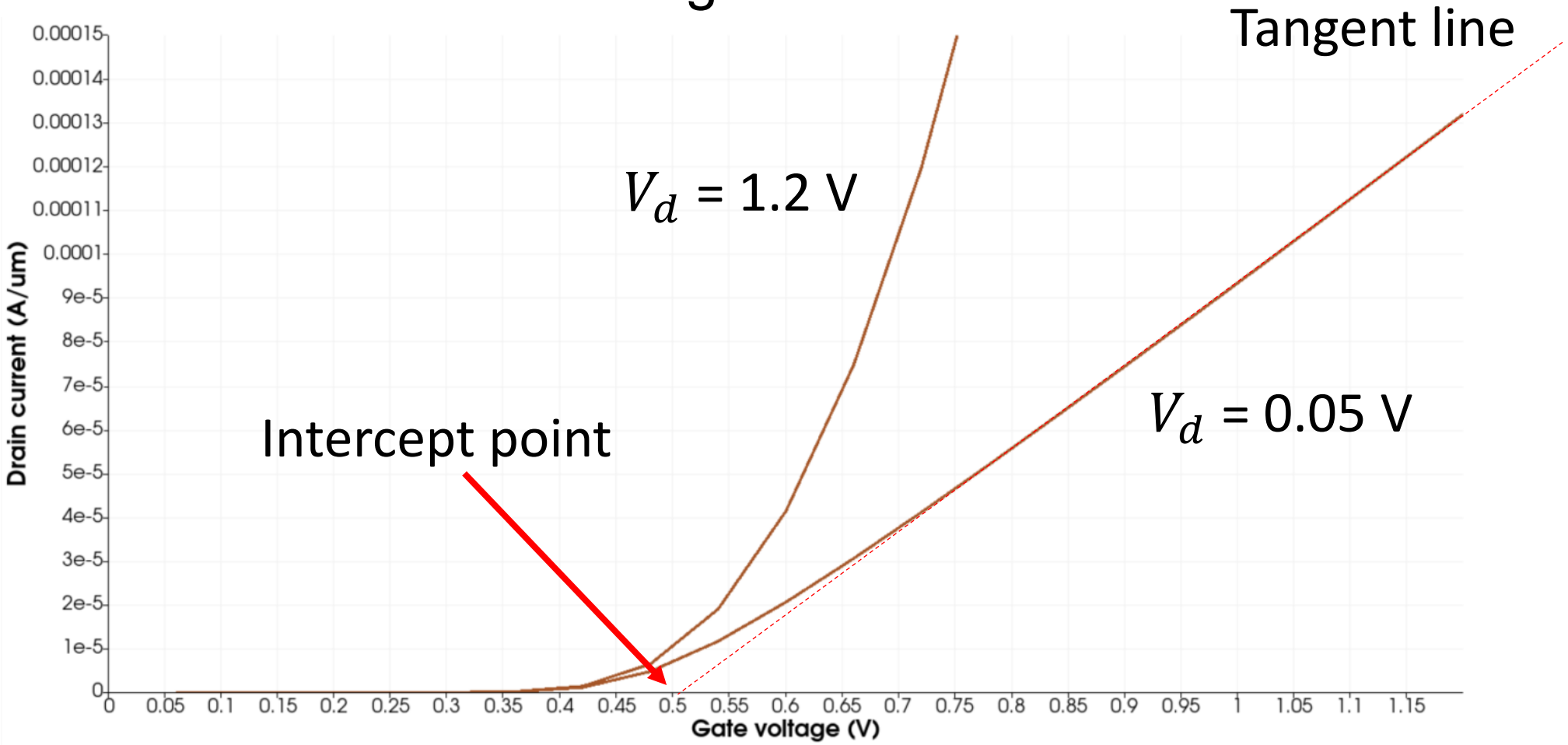
Hole density at $V_g = V_d = 0$ V

- Depletion of holes



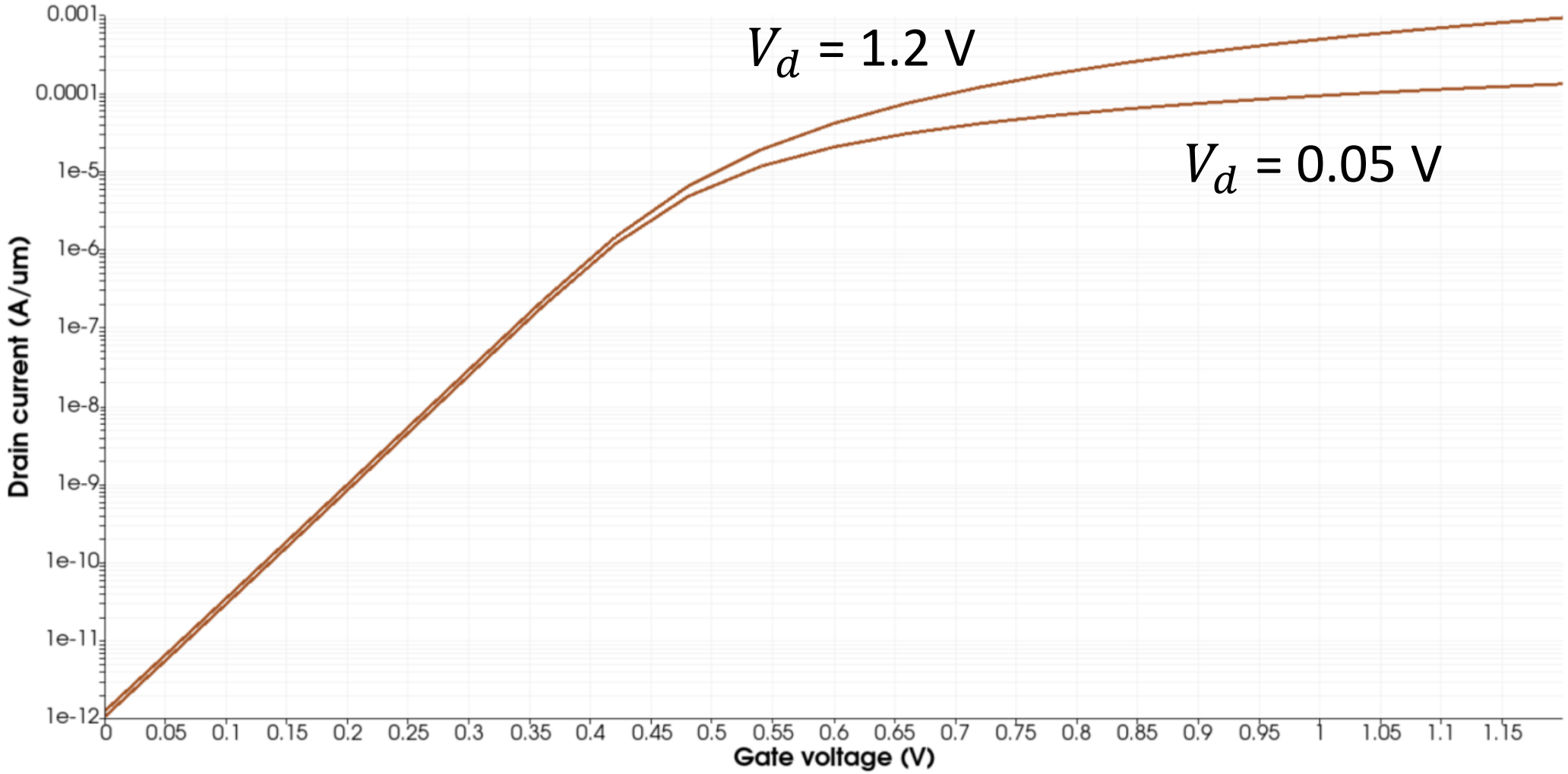
Ramping the gate voltage up

- From 0 V to 1.2 V (V_{dd})
  - Estimate the threshold voltage.



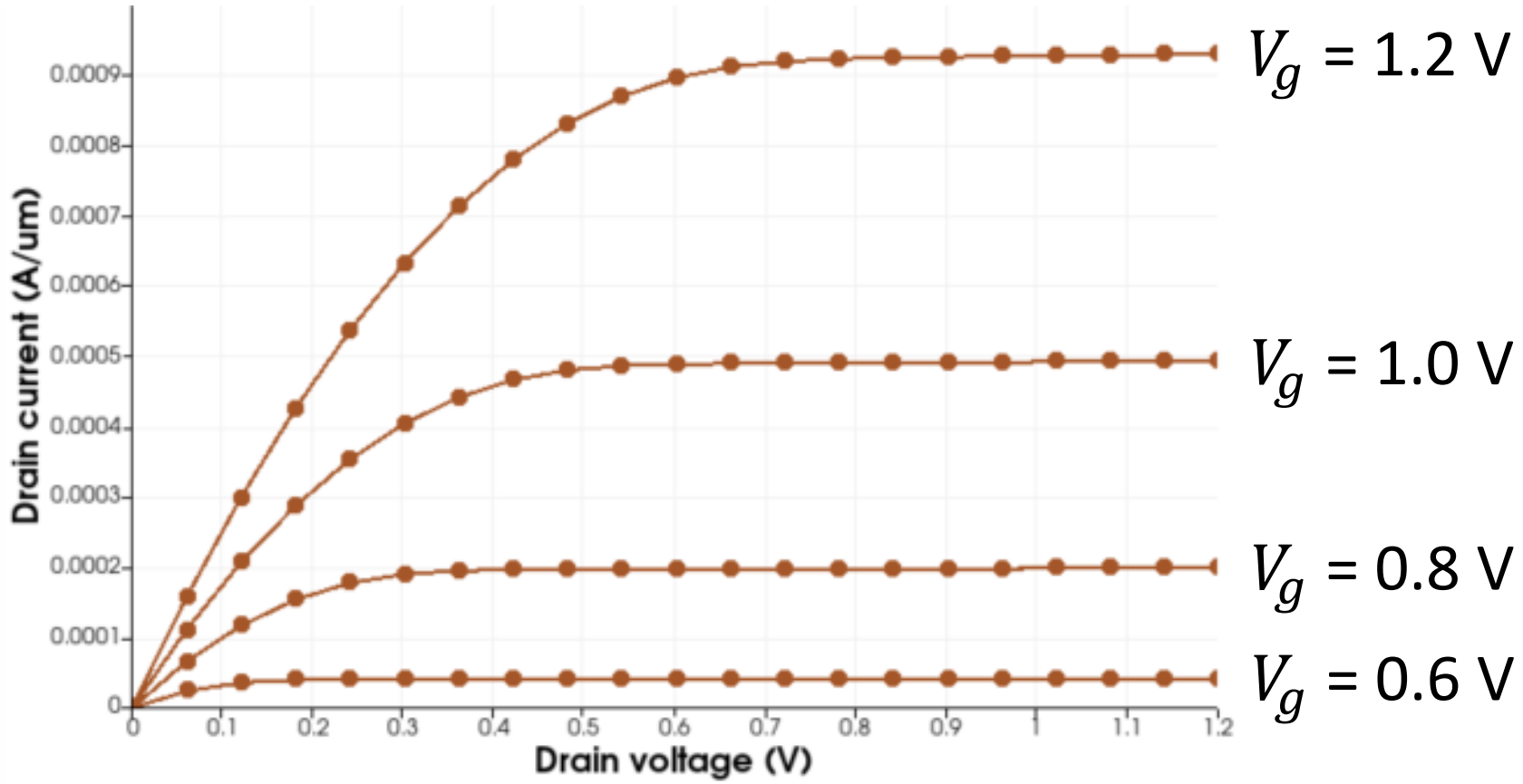
Semi-logarithmic graph

- Estimate the subthreshold slope, the DIBL, I_{ON}, and I_{OFF}.



Output characteristics

- Results from V_g = 0.6 V to 1.2 V (0.2 V step)

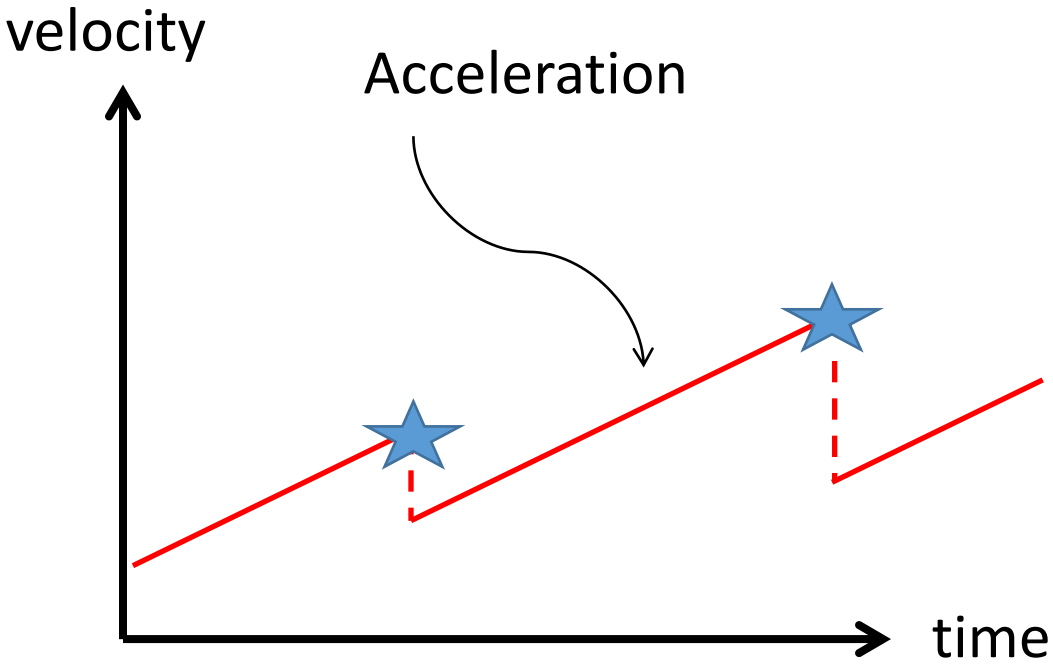


Drift

- Net movement of charge due to an electric field
  - Since electrons/holes are charged particles, they are accelerated by an electric field. (V/cm)
  - For electrons, $\mathbf{F} = -q\mathbf{E}$ (For holes, $\mathbf{F} = +q\mathbf{E}$.)
  - According to Newton's 2nd law, the velocity satisfies $\frac{d\mathbf{v}}{dt} = -\frac{q\mathbf{E}}{m_n}$
  - Here, m_n is the conductivity effective mass of electrons.
  - Then, $\mathbf{v}(t) = \mathbf{v}(0) - \frac{q\mathbf{E}}{m_n}t$ (*Right?*)

Scattering

- The velocity of the carriers…
  - Does not increase indefinitely under the field acceleration. Why?
  - They are scattered frequently and lose the momentum.



Velocity of an electron as a function of time, when scattering events are considered.

Average velocity

- The velocity of each carrier
  - An indivisual electron exhibits sharp transitions.
- The average velocity
  - However, the average velocity follows a much smoother trajectory.
  - Therefore, it would be better to write

$$\frac{d\langle \mathbf{v} \rangle}{dt} = -\frac{q\mathbf{E}}{m_n} - \frac{\langle \mathbf{v} \rangle}{\tau_n}$$

  - Here, τ_n is the mean time between collisions.

Mobility

	μ_n	μ_p
Si	1350	480
GaAs	8500	400
Ge	3900	1900

- Mobility
  - Conduction currents are the result of the drift motion of charge carriers under the influence of an applied electric field.
  - Average drift velocity is directly proportional to the electric field intensity:

$$\langle \mathbf{v} \rangle = -\frac{q\tau_n}{m_n}\mathbf{E} = -\mu_n \mathbf{E}$$

Taur, Eq. (2.31)

Negative sign due to polarity

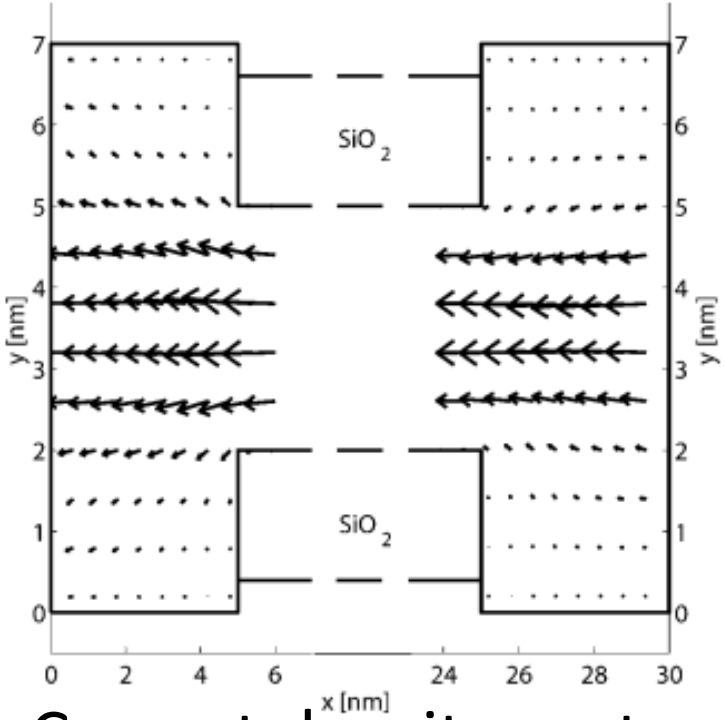
  - μ_n: Electron mobility in ($cm^2\ V^{-1}\ sec^{-1}$)
  - When the above relation is used, the drift current density becomes

$$\mathbf{J} = q\left(\mu_n n + \mu_p p\right)\mathbf{E} = \sigma \mathbf{E}$$

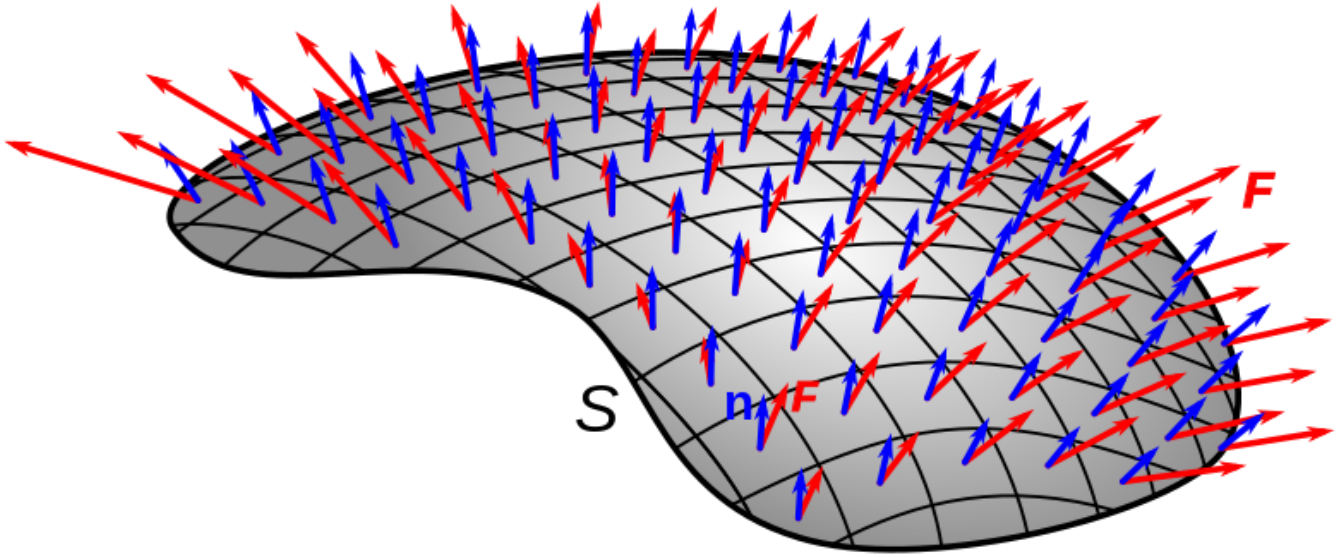
Terminal current vs. current density

- $\mathbf{J}$, current per area (defined everywhere)

$$I_{terminal} = -\int_{terminal} \mathbf{J} \cdot d\mathbf{a}$$



Current density vector (Luisier et al., JAP, vol. 100, p. 043713, 2006)



Surface integral of a vector field (Wikipedia)

Resistivity & sheet resistivity

- Drift current density

$$\mathbf{J}_{n,drift} = q\mu_n n\mathbf{E}$$ Taur, Eq. (2.34)

$$\mathbf{J}_{p,drift} = q\mu_p p\mathbf{E}$$ Taur, Eq. (2.36)

  - Then, the total resistivity is

Ω cm → $$\rho = \frac{1}{q\mu_n n + q\mu_p p}$$ Taur, Eq. (2.38)

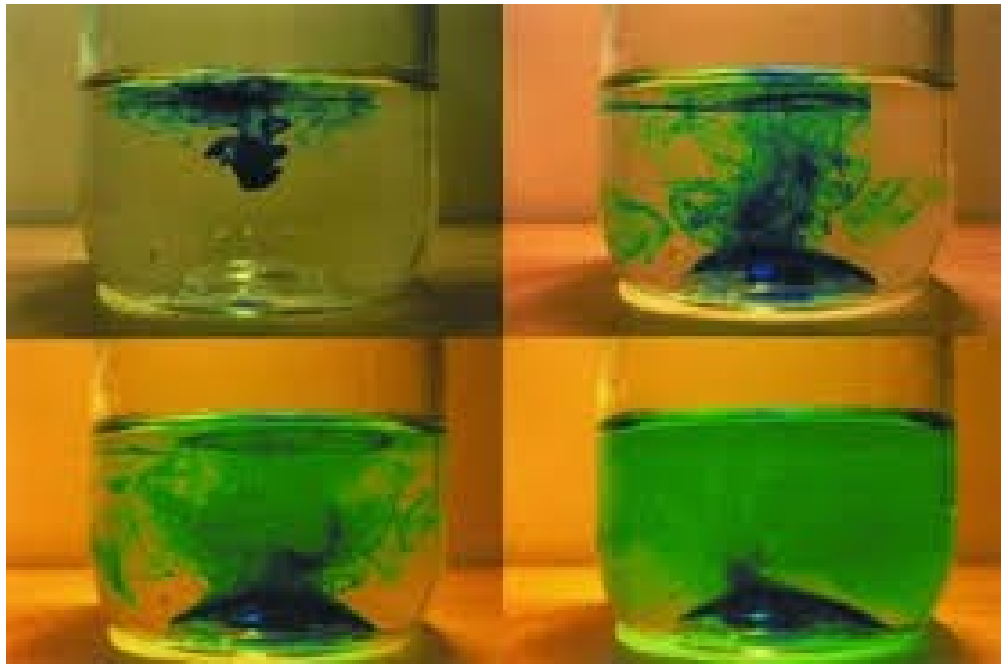
- Sheet resistivity
  - The resistance of a conductor of length, L, and width, W,

$$R = \frac{L}{W}\rho_{sh}$$ Taur, Eq. (2.41)

← Ω/□

Diffusion

- It is not only for charged particles.
  - For example,



Diffusion of ink
(Google images)

  - Therefore, no polarity is expected.

Equation

- Flux
  - The electron flux due to the diffusion mechanism is given by

$$\mathbf{F}_n = -D_n \nabla n$$

  where D_n is the electron diffusion coefficient in the unit of (cm^2/sec).
  - The diffusion current density is

$$\mathbf{J}_{n,diff} = qD_n \nabla n$$

Taur, Eq. (2.46)

- How about the hole?
  - The diffusion current density is

$$\mathbf{J}_{p,diff} = -qD_p \nabla p$$

Taur, Eq. (2.47)

Expressions for current densities

- Total current density
  - Electron current density

$$\mathbf{J}_n = q\mu_n n\mathbf{E} + qD_n\nabla n$$

Taur, Eq. (2.68)

  - Hole current density

$$\mathbf{J}_p = q\mu_p p\mathbf{E} - qD_p\nabla p$$

Taur, Eq. (2.69)

  - Displacement current density

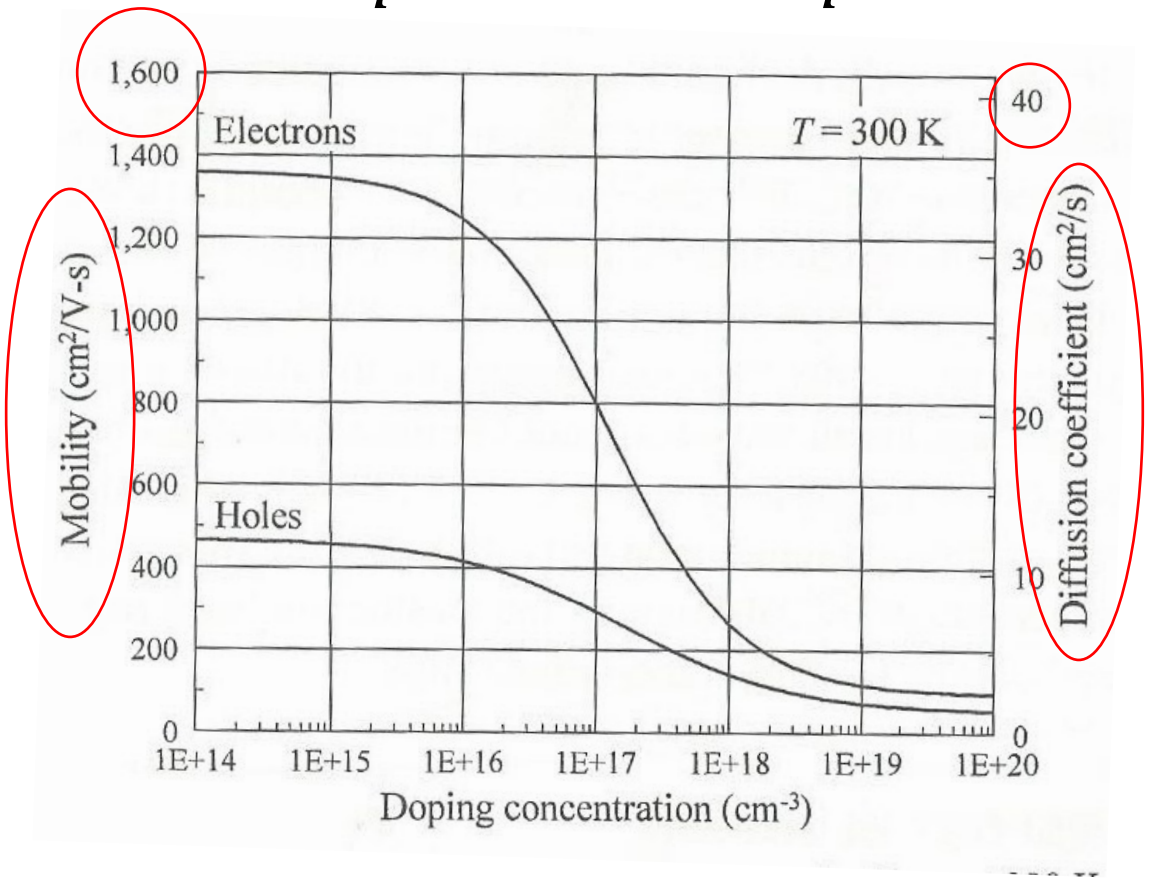
$$\mathbf{J}_{displacement} = \frac{\partial}{\partial t}(\epsilon\mathbf{E})$$

Einstein relation

- At equilibrium, we have the following relations:

$$D_n = \frac{k_B T}{q}\mu_n, D_p = \frac{k_B T}{q}\mu_p$$

Taur, Eq. (2.52) and Eq. (2.53)



(Y. J. Park's book)

Current density with Einstein relation

- Alternative forms
  - Electron current density

$$\mathbf{J}_n = -q\mu_n n\left[\nabla\phi - \frac{k_B T}{q}\frac{1}{n}\nabla n\right]$$

Taur, Eq. (2.70)

  - Hole current density

$$\mathbf{J}_p = -q\mu_p p\left[\nabla\phi + \frac{k_B T}{q}\frac{1}{p}\nabla p\right]$$

Taur, Eq. (2.71)

Revisiting carrier concentrations

$E_f = -q\phi_f$

- Carrier densities are expressed as

$$n = n_i \exp\left(-\frac{E_i - E_f}{k_B T}\right) = n_i \exp\left(\frac{\phi - \phi_f}{k_B T/q}\right)$$ Taur, Eq. (2.63)

$$p = n_i \exp\left(\frac{E_i - E_f}{k_B T}\right) = n_i \exp\left(-\frac{\phi - \phi_f}{k_B T/q}\right)$$ Taur, Eq. (2.64)

  – These relations are generally applicable in the presence of net charge and band bending.

Extension to non-equilibrium cases

- Of course, at non-equilibrium cases, we cannot define E_f.
  - However, we *introduce* ϕ_n and ϕ_p to satisfy:

$$n = n_i \exp\left(\frac{\phi - \phi_n}{k_B T/q}\right)$$ Taur, Eq. (2.75)

$$p = n_i \exp\left(-\frac{\phi - \phi_p}{k_B T/q}\right)$$ Taur, Eq. (2.76)

  - They are called quasi-Fermi potentials. (Electron quasi-Fermi potential and hole quasi-Fermi potential)

Current density with $\nabla\phi_n$ and $\nabla\phi_p$

- From the expression for the quasi-Fermi potential,
  - It can be written as

$$\phi_n = \phi - \frac{k_B T}{q} \log \frac{n}{n_i}$$

Taur, Eq. (2.79)

  - Taking the gradient,

$$\nabla\phi_n = \nabla\phi - \frac{k_B T}{q} \frac{\nabla n}{n}$$

  - Using this expression,

$$\mathbf{J}_n = -q\mu_n n \left[\nabla\phi - \frac{k_B T}{q} \frac{1}{n} \nabla n \right] = -q\mu_n n \nabla\phi_n$$

Taur, Eq. (2.77)

$$\mathbf{J}_p = -q\mu_p p \nabla\phi_p$$

Taur, Eq. (2.78)

Gradient of quasi-Fermi potential

- We have the following relations,

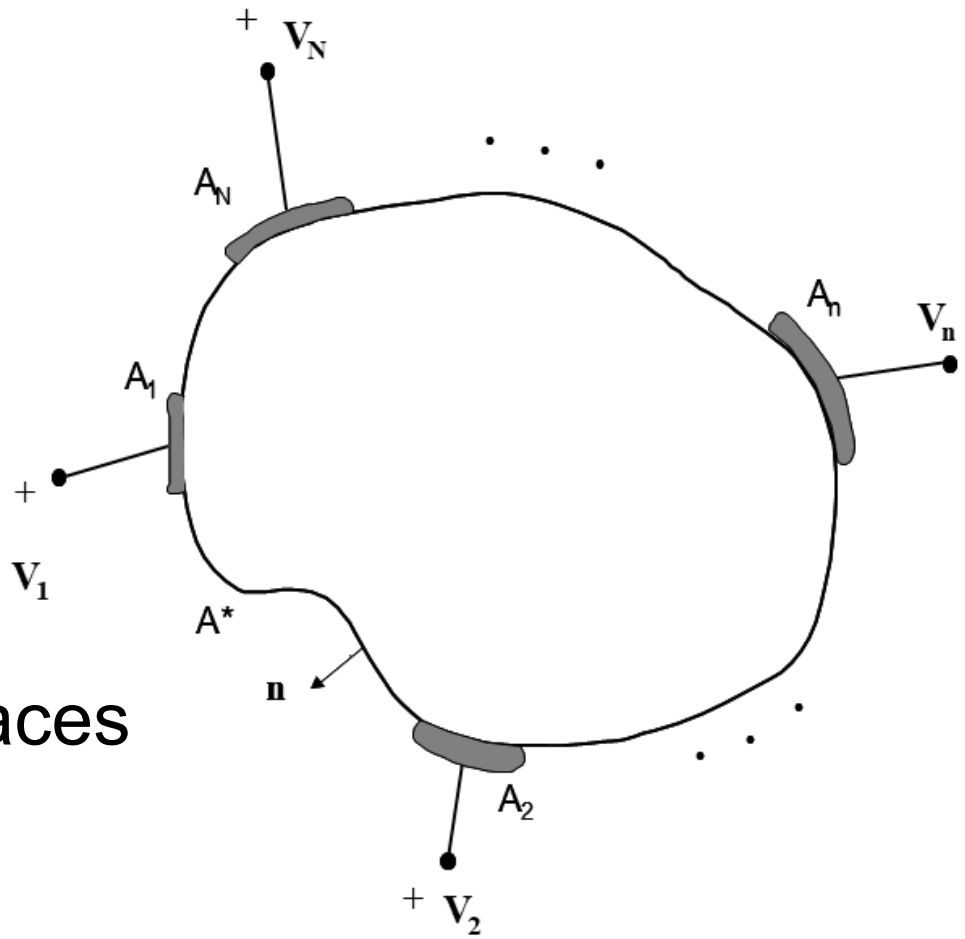
$$\mathbf{J}_n = -\, q\mu_n n \nabla \phi_n$$

$$\mathbf{J}_p = -q\mu_p p \nabla \phi_p$$

  - The gradient of electron quasi-Fermi potential drives the electron current.
  - The gradient of hole quasi-Fermi potential drives the hole current.

Boundary conditions for ϕ_n and ϕ_p

- Dirichlet boundary condition
  - At terminals,
    $\phi_n = \phi_p = V_{app}$



- Neumann boundary condition
  - No current through non-contact surfaces
    $\nabla\phi_n \cdot \mathbf{n} = \nabla\phi_p \cdot \mathbf{n} = 0$

Thank you!